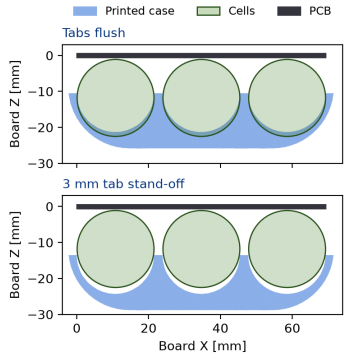
A flush case intersects all three cells before tightening



Flush tabs
The printed case overlaps each cell by about 1.18 mm at its crown. The fit develops contact load during installation.

3 mm stand-off
The unstrained STEP clears the cells and all existing mechanical hardware. The cells can move before reaching the case.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

The October reference gives three quantities to compare

Bare-board quantity	Value	Loading
First prestressed natural frequency	485.964 Hz	Six bolts locked
Maximum PCB directional displacement	0.052674 mm RMS	Base Z
Maximum PCB equivalent stress	90.325 MPa RMS	Base X

The enclosure comparisons will track the corresponding mode shapes, cell-to-case travel, nickel-strip load, PCB stress and the reactions carried by the four case screws.

Geometry and native model preparation are complete. Dynamic and pull results are under study.